

NLP MINI PROJECT

Submitted by:

Vaishnavi Vijayan

24UBC158

3rd BCA A

Public Information Prioritizer Using Natural Language Processing

1. Title

Public Information Prioritizer Using Natural Language Processing

2. Problem Statement

A large amount of public information is generated through news articles, emergency alerts, government announcements, and promotional messages. It can be difficult to identify which information is important and requires immediate attention.

This project develops an NLP-based system that analyzes text and identifies its category, crisis status, crisis type, urgency, action requirement, promotional nature, and final priority level.

3. Objectives

The main objectives of the project are:

- To classify public information into relevant news categories.
- To identify whether a message is related to a crisis.
- To classify the type of crisis when a crisis is detected.
- To estimate the urgency level as Low, Medium, or High.
- To identify whether an action is required.
- To detect promotional or advertisement content.
- To assign a final priority level: Critical, High, Medium, or Low.
- To compare machine-learning models and select the most suitable model based on evaluation results.

4. Dataset

Two datasets were used in this project.

4.1 News Article Categories Dataset

Dataset: news-article-categories.csv

Size: 6,877 records initially; 6,849 records after removing 28 exact duplicate rows.

Attributes:

- Category
- Title
- Body

The dataset contains 14 categories: Arts & Culture, Business, Comedy, Crime, Education, Entertainment, Environment, Media, Politics, Religion, Science, Sports, Tech, and Women.

The title and body were combined to create the input text. Text was converted to lowercase, URLs and mentions were removed, and whitespace was normalized.

4.2 Crisis Dataset

Dataset: crisis.csv

Size: 2,345 records.

Important attributes:

- tweet_id
- user_id
- timestamp
- text
- crisis_type
- location
- relevance_label
- urgency_label
- informative_label

The crisis dataset contains 7 crisis types: Earthquake, Explosion, Fire, Flood, Hurricane, Landslide, and Pandemic.

The dataset contained repeated texts with inconsistent urgency and relevance labels. Therefore, a group-aware split was used for crisis-type classification to prevent the same text from appearing in both training and testing data.

5. Methodology

The overall system follows a hybrid NLP approach:

Input Text → Text Preprocessing → Crisis/Promotion Detection → Classification → Urgency Detection → Action Detection → Priority Assignment

Text Preprocessing

The text is cleaned by:

- Converting text to lowercase.
- Removing URLs.
- Removing mentions.
- Removing unnecessary whitespace.

TF-IDF

TF-IDF (Term Frequency–Inverse Document Frequency) is used to convert text into numerical features based on the importance of words and word combinations.

Machine Learning

Two traditional machine-learning algorithms were experimented with for news-category classification:

- Multinomial Naive Bayes
- Logistic Regression

Logistic Regression was selected because it performed better on the dataset.

For crisis-type classification, TF-IDF with Logistic Regression was used.

Rule-Based NLP

Rule-based techniques were used for:

- Crisis detection
- Urgency detection
- Action-required detection
- Promotional-content detection
- Final priority calculation

This hybrid approach was selected because the available datasets did not provide reliable labels for every required component.

6. Implementation

The project was implemented using Python.

Libraries Used:

- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn
- Joblib
- CSV

Important Techniques

- Text preprocessing
- TF-IDF vectorization
- N-gram features
- Logistic Regression
- Multinomial Naive Bayes

- Rule-based keyword detection
- Stratified train-test splitting
- Group-aware train-test splitting
- Classification report
- Confusion matrix

The news-category model used TF-IDF with unigrams and bigrams and a maximum vocabulary of 20,000 features. The crisis model used TF-IDF with unigrams and bigrams and Logistic Regression.

The final system combines the machine-learning predictions with rule-based decisions to generate a structured result and final priority.

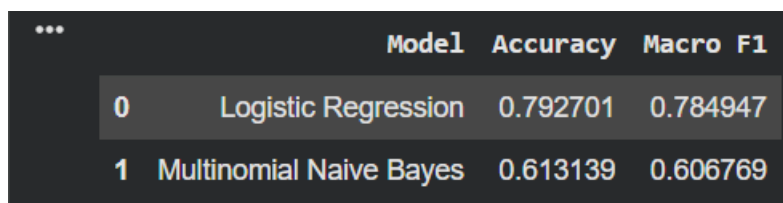
7. Results

7.1 News Category Classification

Two models were evaluated using the same TF-IDF representation.

Model	Accuracy	Macro F1
Multinomial Naive Bayes	61.31%	0.6068
Logistic Regression	79.27%	0.7849

Therefore, Logistic Regression was selected as the final news-category classification model.



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...
      Model Accuracy Macro F1
0  Logistic Regression  0.792701  0.784947
1  Multinomial Naive Bayes  0.613139  0.606769
  
```

Figure 1: Comparison of machine-learning models for news category classification.

7.2 News Category Confusion Matrix

The confusion matrix was used to evaluate the performance of the Logistic Regression model across the 14 news categories. The model performed particularly well for categories such as Sports, Religion, Science, Environment, and Arts & Culture.

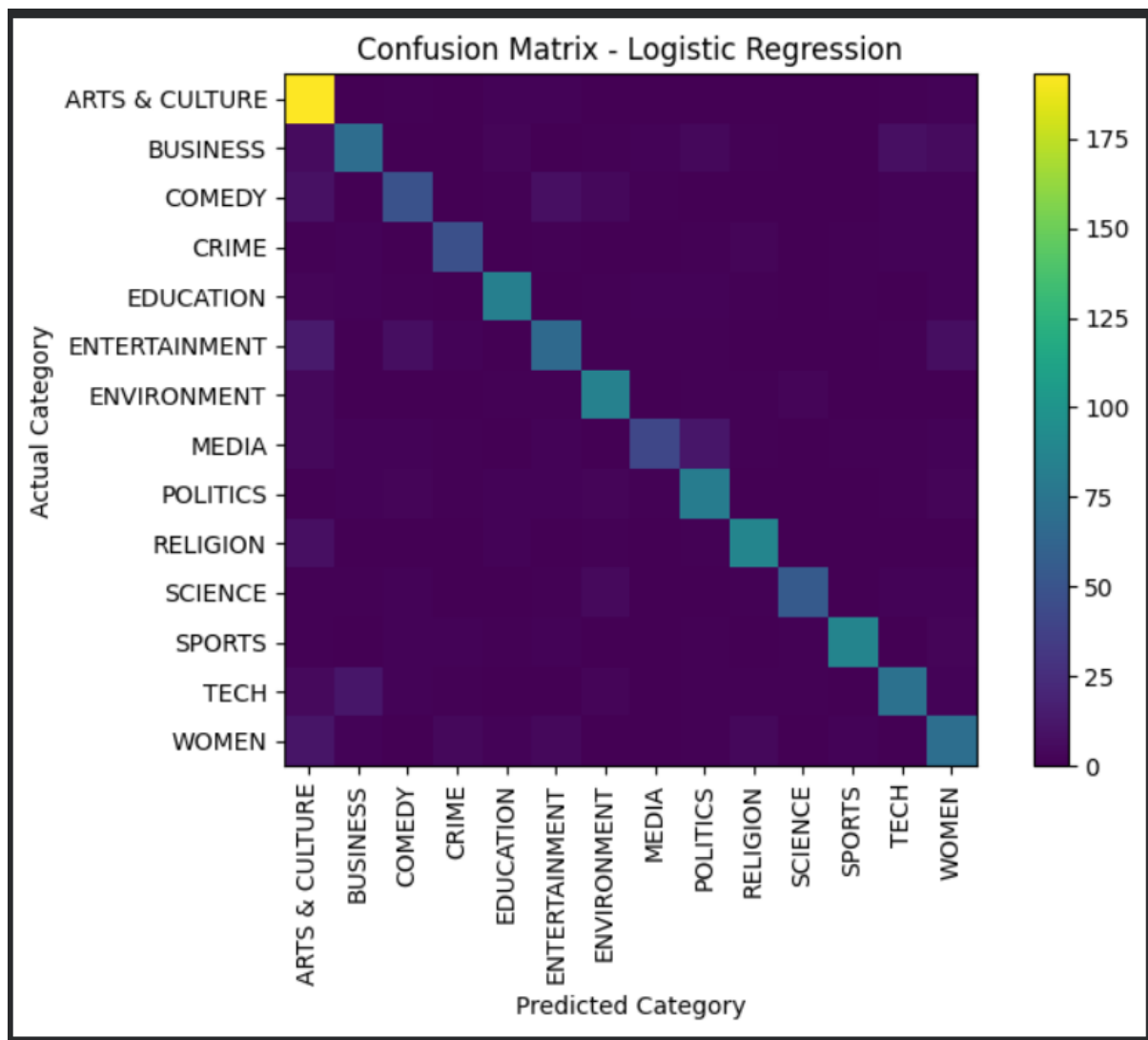


Figure 2: Confusion matrix for news category classification using Logistic Regression.

7.3 Crisis Type Classification

The crisis-type classifier using TF-IDF and Logistic Regression achieved the following results:

Metric	Score
Accuracy	100%
Macro F1	1.00

The model correctly classified all 461 samples in the group-aware test set.

However, the dataset is highly structured, so this result should not be interpreted as guaranteed real-world performance.

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... Crisis Type Accuracy: 1.0

Classification Report:
              precision    recall  f1-score   support

 Earthquake      1.00      1.00      1.00        52
  Explosion      1.00      1.00      1.00       106
    Fire         1.00      1.00      1.00        38
    Flood        1.00      1.00      1.00        84
 Hurricane       1.00      1.00      1.00        39
 Landslide       1.00      1.00      1.00        71
 Pandemic        1.00      1.00      1.00        71

 accuracy                   1.00        461
 macro avg      1.00      1.00      1.00        461
 weighted avg   1.00      1.00      1.00        461

```

Figure 3: Classification report for crisis type prediction.

7.4 Crisis Type Confusion Matrix

The confusion matrix shows that all seven crisis categories were classified correctly in the test set, with no observed misclassifications.

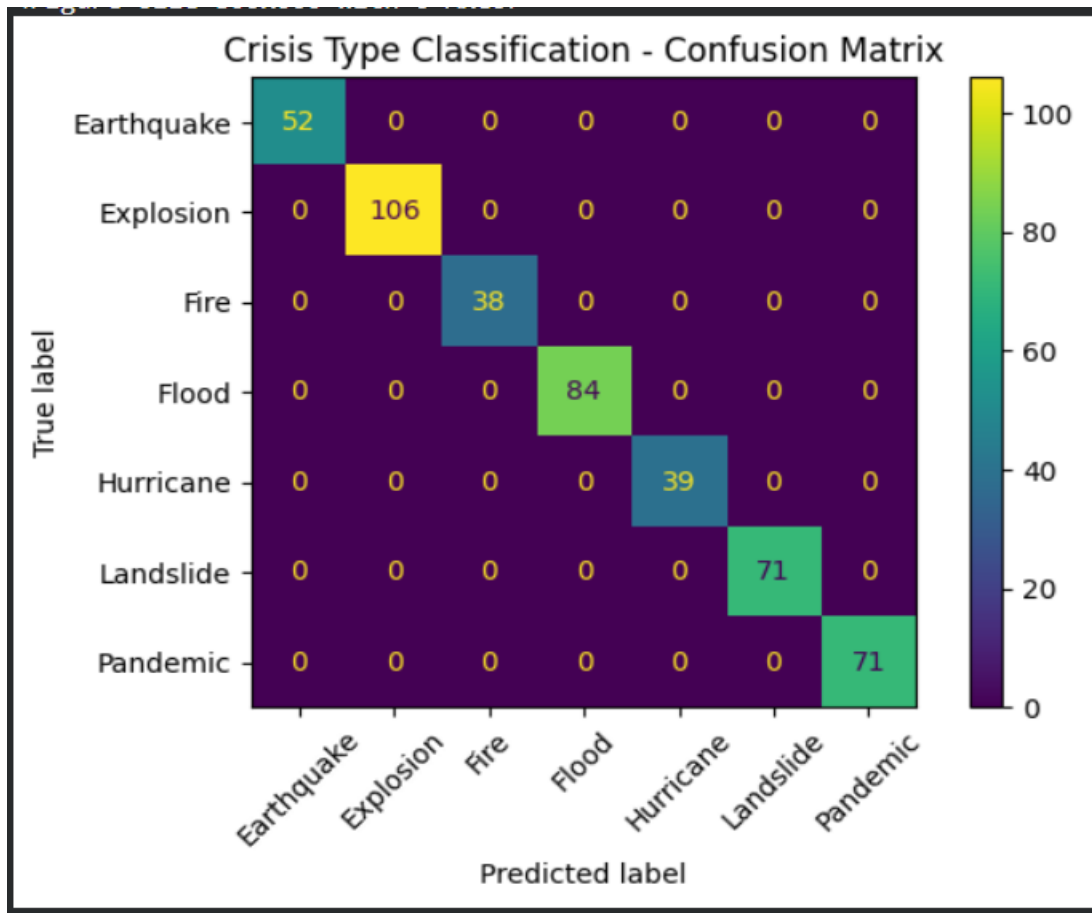


Figure 4: Confusion matrix for crisis type classification using Logistic Regression.

7.5 Urgency Classification Experiment

An ML-based urgency classifier was also experimentally evaluated.

Metric	Score
Accuracy	32.35%
Macro F1	0.2976

Since the performance was approximately at the level expected for a three-class problem and the dataset contained inconsistent urgency labels, the ML urgency classifier was not selected for the final system.

Instead, a transparent rule-based urgency detection method was used.

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... Urgency Accuracy: 0.3235

Classification Report:
              precision    recall  f1-score   support

     High       0.28       0.42       0.33        12
     Low        0.25       0.10       0.14        10
     Medium     0.42       0.42       0.42        12

 accuracy              0.32        34
  macro avg           0.31        0.31       0.30        34
  weighted avg        0.32        0.32       0.31        34

```

Figure 5: Evaluation results of the experimental urgency classifier.

7.6 Final System Sample Output

The final system was tested using different types of public information.

Sample Input

“A powerful earthquake has struck the region. Residents must evacuate immediately.”

Sample Output

Attribute	Prediction
Crisis Status	Crisis
Crisis Type	Earthquake
News Category	N/A
Urgency	High
Action Required	Yes
Promotion	Not Promotional
Final Priority	Critical

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... =====
🔔 PUBLIC INFORMATION PRIORITIZER USING NLP
=====

Enter a news article, alert, notice, or message:
> A powerful earthquake has struck the region. Residents must evacuate immediately.

=====
📊 ANALYSIS RESULT
=====
Crisis Status: Crisis
Crisis Type: Earthquake
News Category: N/A
Urgency: High
Action Required: Yes
Promotion: Not Promotional
Final Priority: Critical
=====
```

Figure 6: Sample output of the final public information prioritization system.

8. Limitations

- The crisis dataset contains many repeated texts.
- Some urgency and relevance labels are inconsistent for repeated texts.
- Rule-based detection depends on predefined keywords.
- The system may not correctly understand complex, ambiguous, or previously unseen language.
- The news-category classifier may confuse semantically similar categories.
- The datasets may not fully represent real-world public information.
- The crisis classifier's 100% test performance may be influenced by the structured nature of the dataset.

9. Future Scope

The system can be improved by:

- Using larger and more diverse datasets.
- Adding multilingual support.
- Using advanced NLP models such as BERT or other transformer-based models.
- Improving urgency and action classification using reliable labelled data.
- Using more advanced semantic understanding instead of keyword-only rules.

- Developing a real-time public information monitoring system.
- Deploying the system as a web or mobile application.
- Integrating the system with news and emergency-alert sources.

10. Conclusion

The project developed a hybrid NLP-based system for analyzing and prioritizing public information. TF-IDF and Logistic Regression were used for text classification, while rule-based NLP was used for urgency, action, promotion, and priority decisions. Logistic Regression achieved 79.27% accuracy for news-category classification and 100% accuracy for crisis-type classification on the evaluated datasets. The project demonstrated how NLP and machine learning can be combined to create a practical information-prioritization system.